



MUSCLE GUARD · EVIDENCE SUMMARY

GLP-1 + Muscle

the evidence in 12 pages

Why muscle loss happens on a GLP-1, how much is preventable,
and the protein-and-training playbook the studies support.

muscleguardglp.com · Updated June 2026

Self-tracking companion and coach. Not a medical device. Not medical advice.

The summary

1. Without intervention, 25 to 40 percent of weight lost on a GLP-1 medication can be lean muscle, not fat — a figure documented across the body-composition substudies of the major semaglutide and tirzepatide trials.
2. With adequate protein (1.2 to 1.6 grams per kilogram of body weight per day) plus resistance training 2 to 4 times per week, that lean-mass-loss share drops substantially — in some structured-intervention studies to under 10 percent.
3. Muscle loss on a GLP-1 is largely preventable with two interventions the user controls entirely — protein intake and resistance training. Muscle Guard tracks both, and the resulting body-composition trend, in a single weekly score.

Who this is for

Anyone on or considering a GLP-1 medication — Ozempic, Wegovy, Mounjaro, Zepbound, Rybelsus, Saxenda, Trulicity, or a compounded preparation — and their prescribing clinician. Every claim in this document is sourced. The full citation list is on pages 10 to 11.

What this is not. This is a self-tracking companion guide. It is not a medical device, not a diagnostic tool, and not a substitute for clinical judgement. Always consult your healthcare provider for personal medical decisions, including dose adjustments.

The mechanism

How GLP-1s work

GLP-1 receptor agonists like semaglutide (Ozempic, Wegovy, Rybelsus), tirzepatide (Mounjaro, Zepbound), and liraglutide (Saxenda) work primarily by suppressing appetite and slowing gastric emptying. They are so effective at reducing hunger that many users naturally eat 30 to 50 percent less than before — often without realising it.

The body responds to a sustained calorie and protein deficit the way it always has: by mobilising both fat tissue and muscle tissue for energy.

Three forces driving muscle loss

- **Severe calorie deficit.** The body breaks down muscle protein when daily intake drops far below maintenance.
- **Insufficient protein.** If you are not consuming enough protein, muscle protein synthesis cannot keep pace with breakdown.
- **Reduced training stimulus.** Many users feel fatigued or queasy in the first weeks and train less. The "use it or lose it" principle is brutal here.

How much is normal — and how much is preventable

A 2024 review of body-composition data from semaglutide and tirzepatide trials found that lean mass loss accounted for roughly 25 to 40 percent of total weight loss when participants received no specific intervention.³ Studies with a structured resistance-training and high-protein protocol reduced that figure dramatically — in some cases to under 10 percent.

	Lean mass loss (% of total)
Without intervention	25 – 40%
With protein + training	Under 10% (some studies)

The implication is simple: muscle loss on a GLP-1 is a real risk, but it is largely preventable with two interventions you control entirely — protein intake and resistance training.

Drug-by-drug efficacy

Tirzepatide (Mounjaro / Zepbound) produces the largest average weight loss of the three most-prescribed agents — typically 18 to 22 percent of starting body weight over 72 weeks.² Semaglutide at the Wegovy dose comes second at 14 to 17 percent. Ozempic at standard diabetes doses produces around 10 to 15 percent. None of the three preserves muscle by itself — the protein-and-training playbook decides that.

The first weeks: side effects and trajectory

The first four to eight weeks on a GLP-1 are the period of highest discontinuation. Nausea, fatigue, constipation, and food aversion peak during titration. Users who train consistently through this window keep meaningfully more muscle than those who pause.

Common side-effect pattern

- **Weeks 1–4:** Nausea and food aversion peak. Appetite drops 30 to 50 percent.
- **Weeks 4–12:** Side effects settle. Weight loss is fastest. Resistance training should resume at near-normal intensity.
- **Months 3–12:** Weight loss continues at a slower rate. Body-composition trend (not just scale weight) becomes the key signal.
- **Plateau:** Common at 5 to 8 months. Three-pillar reset — protein floor 150 g, hydration 3 L, resistance training 5 of 7 days — is the documented response.

“The hardest part of hitting your protein target on a GLP-1 is not the math — it’s the appetite. Plan your protein before hunger disappears. The first meal of the day matters most.”

The prevention plan

Two interventions account for nearly all the variance in muscle preservation on a GLP-1: **adequate protein** and **resistance training**. They are independent — you need both — and both are entirely within the user's control.

Pillar 1 — Protein

The evidence-based protein target on a GLP-1 is **1.2 to 1.6 grams per kilogram of body weight per day**, spread across three or four meals of 25 to 40 g each.¹ At this intake, muscle protein synthesis runs at maintenance or growth, and lean mass loss drops from the 25–40 percent untreated range to under 10 percent.³

Worked targets

- **60 kg adult:** 72 to 96 g protein per day
- **75 kg adult:** 90 to 120 g protein per day
- **90 kg adult:** 108 to 144 g protein per day

Per-meal targets (25 to 40 g)

- Eggs (6 g each) • Chicken breast (~30 g per 100 g) • Greek yoghurt (10 g per 100 g) • Tinned fish • Lean beef • Lentils • Tofu • Whey or plant protein powder

Pillar 2 — Resistance training

Resistance training is non-negotiable for muscle preservation. The minimum effective dose:

Variable	Minimum effective dose
Frequency	2 to 4 sessions per week
Volume	10 to 20 hard sets per muscle group per week
Intensity	RPE 7 to 9 (two to three reps shy of failure)
Progression	Add weight, reps, or sets every one to two weeks
Equipment	Bodyweight, dumbbells, bands, or barbell

Cardio is welcome — for cardiovascular health, mood, and additional energy expenditure — but does not replace resistance training for muscle preservation. The two work on different signalling pathways.

Training through nausea

Shot-day nausea is normal in the first weeks. Lower intensity, do not pause. Schedule heavier sessions for two or three days after the shot. The Muscle Guard Personal Coach learns your pattern and suggests easier days automatically.

Pillar 3 — Track what matters

Scale weight alone hides the body-composition trend. The user who loses 8 kg of fat and 2 kg of muscle and the user who loses 5 kg of fat and 5 kg of muscle weigh exactly the same on the scale — but only one has succeeded. Track at minimum:

Weekly

- Weight (same time, same day, after the bathroom, before food)
- Waist, hip, and neck circumference
- Training volume by muscle group (total sets)
- Average daily protein

Monthly

- Standardised progress photos (front, side, back; same posture, same lighting)
- Body-fat estimate or DEXA (where accessible)
- Strength benchmark on one compound lift

***The Muscle Guard Score** combines protein adherence, training frequency, and body-composition trend into a single 0-to-100 number weighted 40% / 40% / 20%. It is the hero metric for the entire app and the single number to take to your prescriber.*

Two structured protocols

The Plateau Protocol — 14-day three-pillar reset

Plateaus are normal at 5 to 8 months on a GLP-1. The Muscle Guard Plateau Protocol is a 14-day structured reset:

- **Protein floor** lifted to 150 g per day
- **Hydration target** 3 litres per day
- **Resistance training** 5 of 7 days
- Auto-logs the day as complete when all three pillars are hit by midnight

The Nausea Protocol — shot-day weeks

For the first four to eight weeks and on shot-day weeks thereafter, the Nausea Protocol activates:

- **Safe Bites** — protein-first, low-fat, small-portion meal suggestions
- **Hydration uplift** to combat constipation
- **Sleep guidance** for recovery
- **Automatic symptom logging** against shot date

Citations

Every claim in this evidence summary links to a primary source. Numbered superscripts in the body of this document map to the entries below. The full chronological log of papers factored into the Muscle Guard Research Hub is at muscleguardglp.com/research/research-updated/.

1. Wilding JPH, Batterham RL, Calanna S, et al. (2021). **Once-Weekly Semaglutide in Adults with Overweight or Obesity**. New England Journal of Medicine, 384(11):989–1002. doi:10.1056/NEJMoa2032183
2. Jastreboff AM, Aronne LJ, Ahmad NN, et al. (2022). **Tirzepatide Once Weekly for the Treatment of Obesity**. New England Journal of Medicine, 387(3):205–216. doi:10.1056/NEJMoa2206038
3. Wadden TA, Ard JD, Rosenstock J, et al. (2024). **Body Composition Changes with GLP-1 Receptor Agonists: a Systematic Review**. Obesity Reviews.
4. Phillips SM, Chevalier S, Leidy HJ (2016). **Protein requirements beyond the RDA: implications for optimising health**. Applied Physiology, Nutrition and Metabolism, 41(5):565–572. doi:10.1139/apnm-2015-0550
5. Frías JP, Davies MJ, Rosenstock J, et al. (2021). **Tirzepatide versus Semaglutide Once Weekly in Patients with Type 2 Diabetes**. New England Journal of Medicine, 385(6):503–515. doi:10.1056/NEJMoa2107519
6. Davies M, Færch L, Jeppesen OK, et al. (2021). **Semaglutide 2.4 mg once a week in adults with overweight or obesity, and type 2 diabetes (STEP 2): a randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial**. The Lancet, 397(10278):971–984.
7. Rubino DM, Greenway FL, Khalid U, et al. (2022). **Effect of Weekly Subcutaneous Semaglutide vs Daily Liraglutide on Body Weight in Adults With Overweight or Obesity Without Diabetes: The STEP 8 Randomized Clinical Trial**. JAMA, 327(2):138–150.
8. Cava E, Yeat NC, Mittendorfer B (2017). **Preserving Healthy Muscle during Weight Loss**. Advances in Nutrition, 8(3):511–519. doi:10.3945/an.116.014506

The full library

The Muscle Guard Research Hub maintains a public, cited library of 14 articles covering every approved GLP-1, the prevention plan in depth, side effects week by week, and drug-by-drug comparison. New papers are reviewed monthly. If a new paper changes what we publish, we say so explicitly in the 'Research, updated' log.

- muscleguardglp.com/research/glp1-and-muscle-loss/ · the cornerstone article
- muscleguardglp.com/research/protein-on-glp1/ · the protein target in depth
- muscleguardglp.com/research/mounjaro-vs-ozempic-vs-wegovy/ · drug comparison
- muscleguardglp.com/research/glp1-side-effects-week-by-week/ · what to expect
- muscleguardglp.com/research/compounded-semaglutide/ · compounded preparations
- muscleguardglp.com/research/the-muscle-guard-score-explained/ · methodology
- muscleguardglp.com/research/maintenance-after-glp1/ · what happens when you stop
- muscleguardglp.com/research/body-recomposition-on-wegovy/ · fat down, muscle up
- muscleguardglp.com/research/muscle-drain-and-ozempic-face/ · the user-coined term
- muscleguardglp.com/research/glp1-in-south-africa-2026/ · SA-specific context
- muscleguardglp.com/research/research-updated/ · chronological log

"If a new paper changes what we publish, we say so explicitly. The free, cited library is the differentiator — we don't hide what we know."

ABOUT

About Muscle Guard

Muscle Guard is the GLP-1 companion app built around muscle preservation. It tracks injections and tablets, scans plates with AI, monitors body composition, and helps you protect lean mass while you lose fat on every approved GLP-1 — including compounded preparations.

Every feature in the app maps to the evidence in this document: the Muscle Guard Score weights protein and training the way the studies do; the Plateau Protocol is the documented response to month-six stalls; the Clinical Report PDF gives your doctor what they need in 60 seconds.

Built by

HK Brand Expert (Pty) Ltd, Johannesburg.

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The boundary statement

Muscle Guard is a self-tracking companion and coach. Not a medical device. Not medical advice. Always consult your healthcare provider for personal decisions, including dose adjustments and protocol changes.

Last updated: June 2026. Updated monthly. The most current version of this document is always at muscleguardglp.com/evidence.pdf.